# Relatives' similarity in test score performance is explained by a wide range of genetic and environmental parameters

Supplemental slides

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Reanalysis of Eftedal NH, Eilertsen EM, Sunde HF, Kleppestø TH, Ystrom E, Czajkowski NO (2025). *PNAS* 122(25):e2419627122.

github.com/nunorgcarvalho/eftedal_reanalysis nuno_ramosgrangeiacarvalho@g.harvard.edu

# AI Disclosure

I used Claude (Opus 5.5) extensively to help me with this project, and almost everything in this repository was ultimately written by Claude. However, I was responsible for deciding which analyses to run and was constantly in the loop regarding their implementation and provided direct feedback about improving it.

With the exception of this disclosure, this README was written by Claude, as well as the figures, tables, poster(s), slides, notebooks, and more. However, the text on the poster and many of the slides here was written by me.

— from the repository README

# Clarifications to the poster since it was printed

- **The A + GxG + C model was chosen as the example sicne it ranks first in our cross-validation ranking** among the core models (slides 29–30): its CV RMSE is 0.035 vs 0.070 for the Additive model (0.030 vs 0.055 if adoptive relatives are removed, since the additive model predicts them always to be 0). However, the improvement is small (difference is about 1.7 standard errors).
- **Component estimates within a model are collinear,** which can be visualized in the leave-one-type-out jackknife correlation plots shown below.
- **Marginal Type weights ("uniform across types"):** all sorts of specific relationship types were condensed into 10 broad relative blocks. Each block gets the same total weight, and the block's pooled correlation is computed from inverse-variance $(1/\mathrm{SE}^2)$ weights applied to relationship types within a block.
- **"All relative types" in the Additive-model inset:** in Eftedal et al., their excluded adoptive types and twins of unknown zygosity from their fits (due to their estimator not working for pairs with an expected covariance of 0). In our analyses, we include all relative types when estimating paramters.

# The core models

how to read the model slides

### Components

- **A**: additive genetic effects under assortative mating (AM); the model of Eftedal et al.
- **GxG**: pairwise gene–gene interactions, decaying as $(1/4)^n$ with genetic degree $n$ (not affected by AM).
- **C**: shared environment, $V_C\, c^f$ with family degree $f$; split into a within-sibling part $V_C(1-c)$ and a beyond-sibling part $V_C\, c$.

### Naming

- **+** : components fitted jointly.
- $\rightarrow$ : fitted in steps. The components before the arrow are fitted first and their parameters are then held fixed while those after the arrow are fitted.

### Model slides

Revised estimator, optimized for the Marginal Type, as on the poster. Marginal Pair CIs: parametric bootstrap; Marginal Type CIs: leave-one-type-out jackknife. CV: leave-one-type-out cross-validation (RMSE; rank among the nine models, 1 = best).

| symbol | meaning |
|---|---|
| *Additive model* | |
| $n$ | genetic degree |
| $V_A$ | additive genetic variance |
| $V_A^0$ | additive variance under random mating, $V_A(1-m)$ |
| $m$ | mates' genetic correlation |
| $r$ | mates' phenotypic correlation |
| *Gene–gene interactions* | |
| $V_{\mathrm{GxG}}$ | GxG variance |
| *Shared environment* | |
| $V_C$ | shared-environment variance |
| $f$ | family degree |
| $c$ | decay per family degree |

Section 1

# Revising Eftedal et al.’s results

# How Eftedal et al. fit their model (1/2)

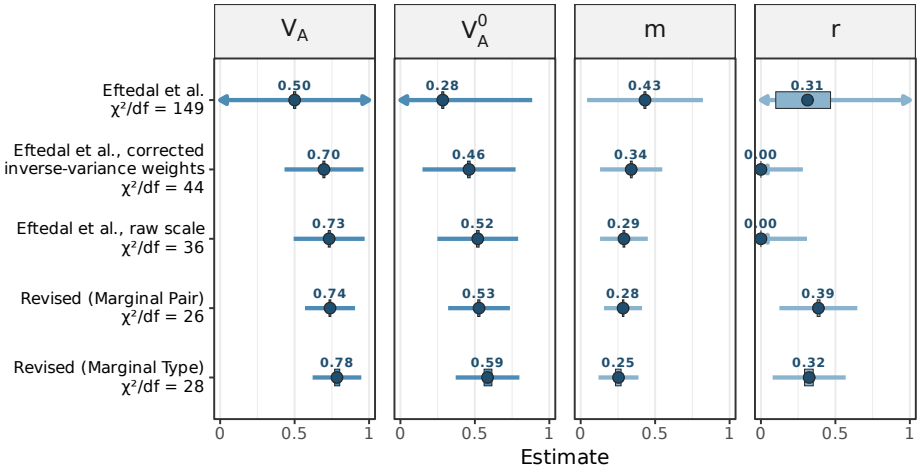
| Eftedal et al.'s decision | Consequence | Our fix |
|---|---|---|
| Identified 82 relationship types, split by type of relative, sex of the pair, and the side of the family through which the pair is related. | Distant relatives have a greater number of "distinct" relationship types. | Grouped the relationship types into 10 broad types of relatives. |
| Minimized the residual sum of squares (RSS; squared difference between predicted and observed correlation, summed across relationship types). | Distant relative types (with more "distinct" relationships) make up a greater share of the RSS. | No change. |
| Optimized for log-transformed correlations, which makes optimization a simpler linear regression problem. | Squared errors are greater for smaller raw correlations, so distant relatives make up an *even greater* share of the RSS. | Optimized for observed correlations on the raw scale, using non-linear optimization. |
| Weighted each relationship type's residual by the inverse variance of its observed correlation. | Effectively optimizes for the marginal pair of individuals in the full population, which is overwhelmingly weighted towards distant relatives (much more numerous, with lower observed correlations). Distant relatives make up an *even greater* share of the RSS. | Fit for the Marginal Type instead: each of the 10 relative types gets equal total weight; within a type, relationship types are weighted by inverse variance. |

# How Eftedal et al. fit their model (2/2)

| Eftedal et al.'s decision | Consequence | Our fix |
|---|---|---|
| Inverse-variance weights based on the raw correlation ($1/\mathrm{SE}^2$), not the log-correlation ($\approx \rho^2/\mathrm{SE}^2$). | Distant relatives (who have low correlations) make up an *even greater* share of the RSS. | Confirmed that the correct inverse-variance weights on the log scale give estimates similar to optimizing on the raw scale. |
| Used the wrong expected correlation for step-siblings, $V_A\, m\, \big((1+m)/2\big)^2$. The primary assortative mating model they assume throughout instead predicts $V_A\, m\, \big((1+r)/2\big)^2$. | Predicts step-siblings to be less similar than the model actually implies (since $r > m$). | Used the correct formula. |
| Estimated $m$ and $r$ independently, despite the primary assortative mating model implying $m = V_A\, r$. | Mismatch between the estimator and the estimand. | Enforced the constraint implied by the model. |
| Confidence intervals from bootstrap re-sampling of the observed correlations (relationship types drawn with probability $\propto 1/\mathrm{SE}^2$), using the same transformation and weights as the point estimates. | The same biases as the point estimates (fits overwhelmingly to distant relatives), and overly confident intervals. | Confidence intervals from a leave-one-type-out jackknife over the 10 relative types. |

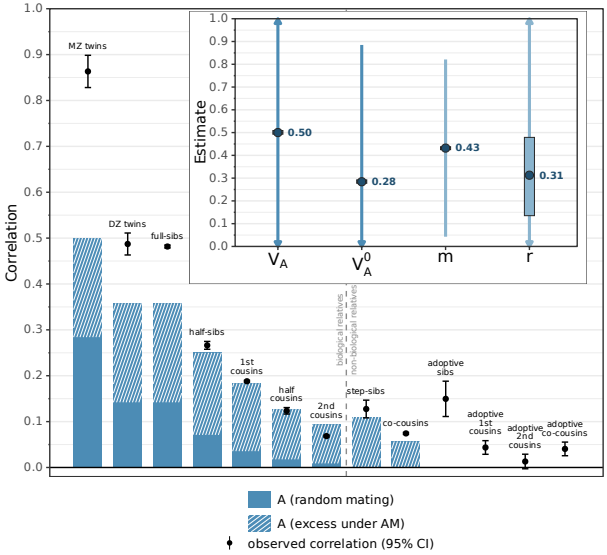
# From Eftedal et al.'s estimator to the revised one

additive model




Row 2: log scale with the correct inverse-variance weights ($\approx \rho^2/\mathrm{SE}^2$). Row 3: raw scale, where $1/\mathrm{SE}^2$ is correct. Rows 4–5: all fixes (raw scale, $m = V_A\, r$, corrected step-sibling formula, all 82 relationship types). Circle: estimate; box: Marginal Pair CI; line: Marginal Type CI (arrow: truncated at 0 or 1). $\chi^2/\mathrm{df}$ over each fit's own relationship types.

# Eftedal et al. (A) Marginal Pair fit




## Model

The additive model with Eftedal et al.'s estimator and settings, fitted to biological relatives and in-laws.

**Fit** (Marginal Pair)

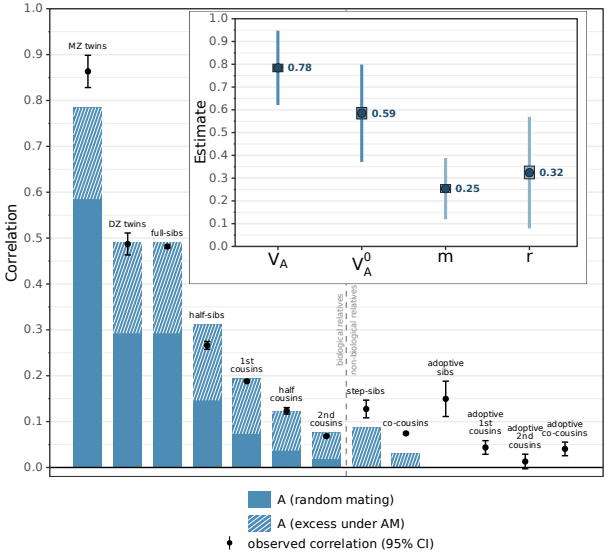
$\chi^2/\mathrm{df}$ 149.4

## Comments

- Under-predicts twins and siblings while fitting distant relatives closely.
- $\chi^2/\mathrm{df}$ over the relative types they fitted (no adoptive types).

Bars: predicted, by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line; arrow: truncated at 0 or 1).

# Revised (A) Marginal Type fit




## Model

Additive genetic effects under assortative mating only.

**Fit** (Marginal Type)

| | |
|---|---|
| $\chi^2/\mathrm{df}$ | 28.1 |
| CV RMSE | 0.070 |
| CV rank | 7 of 9 |

## Comments

- The model used in Eftedal et al., but fitted for the marginal relative type rather than the marginal pair of individuals.
- The additive component (in the current and all subsequent models) has only two degrees of freedom, as $V_A^0 = V_A(1-m)$ and $m = V_A\, r$.

Bars: predicted, by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line; arrow: truncated at 0 or 1).

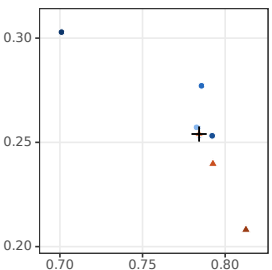
# Revised (A) parameter correlations, Marginal Type fit

$V_A$

0.784

[0.621, 0.948]

r = -0.85




m

0.254

[0.120, 0.388]

Refit leaving out

- twins
- full-sibs
- half-sibs
- 1st cousins
- half cousins
- 2nd cousins
- step-sibs
- co-cousins
- adoptive sibs
- adoptive cousins

## Reading the plot

**Lower:** estimates from the 10 leave-one-type-out refits (colour: type left out); cross: full-data estimate.

**Upper:** $r$, the correlation of the refit estimates (jackknife sampling correlation).

**Diagonal:** estimate and Marginal Type CI.

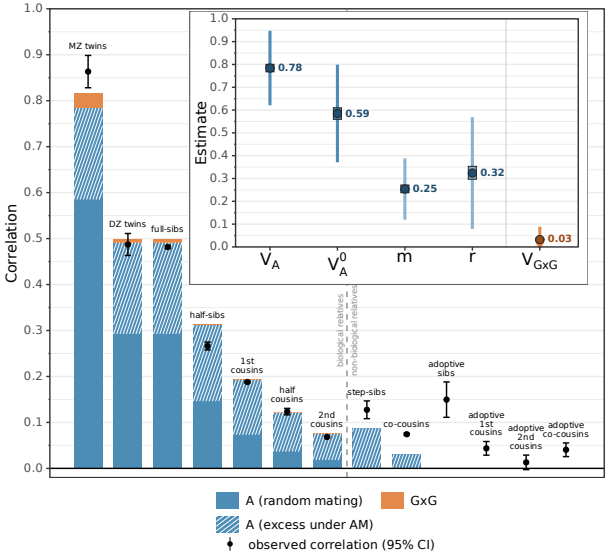
## Comments

- $V_A$ and $m$ trade off strongly ($r = -0.85$).

With only 10 relative types, one or two influential types can set $r$: read it as the direction of the trade-off.

Section 2

# Models with GxG and shared environment



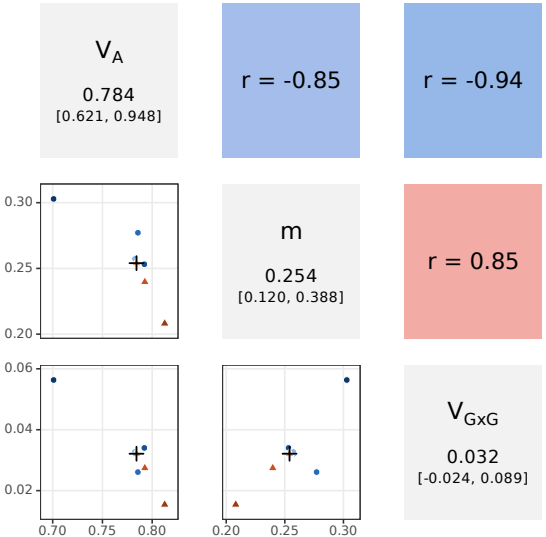

## Model

**A** fitted first; **GxG** then fitted to what remains, with **A**'s parameters held fixed.

**Fit** (Marginal Type)

| | |
|---|---|
| $\chi^2/\mathrm{df}$ | 29.5 |
| CV RMSE | 0.065 |
| CV rank | 5 of 9 |

Bars: predicted, by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line; arrow: truncated at 0 or 1).




Refit leaving out

- twins
- full-sibs
- half-sibs
- 1st cousins
- half cousins
- 2nd cousins
- step-sibs
- co-cousins
- adoptive sibs
- adoptive cousins

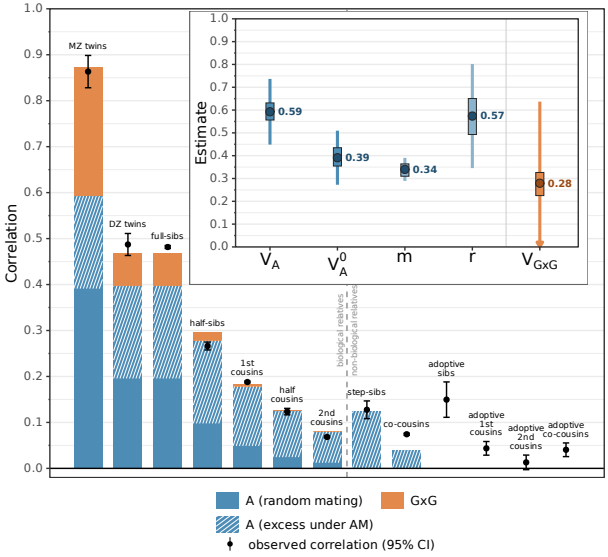
## Reading the plot

**Lower:** estimates from the 10 leave-one-type-out refits (colour: type left out); cross: full-data estimate.

**Upper:** $r$, the correlation of the refit estimates (jackknife sampling correlation).

**Diagonal:** estimate and Marginal Type CI.

With only 10 relative types, one or two influential types can set $r$: read it as the direction of the trade-off.



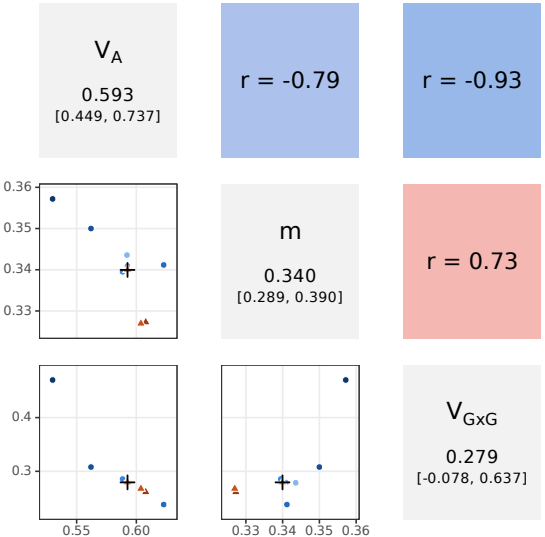

## Model

**A** and **GxG** fitted jointly.

**Fit** (Marginal Type)

| | |
|---|---|
| $\chi^2/\mathrm{df}$ | 21.2 |
| CV RMSE | 0.063 |
| CV rank | 4 of 9 |

Bars: predicted, by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line; arrow: truncated at 0 or 1).




Refit leaving out

- twins
- full-sibs
- half-sibs
- 1st cousins
- half cousins
- 2nd cousins
- step-sibs
- co-cousins
- adoptive sibs
- adoptive cousins

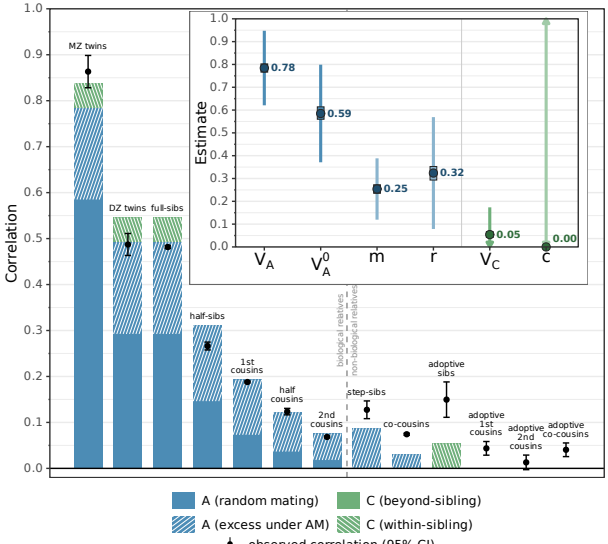
## Reading the plot

**Lower:** estimates from the 10 leave-one-type-out refits (colour: type left out); cross: full-data estimate.

**Upper:** $r$, the correlation of the refit estimates (jackknife sampling correlation).

**Diagonal:** estimate and Marginal Type CI.

With only 10 relative types, one or two influential types can set $r$: read it as the direction of the trade-off.




## Model

**A** fitted first; **C** then fitted to what remains, with **A**'s parameters held fixed.

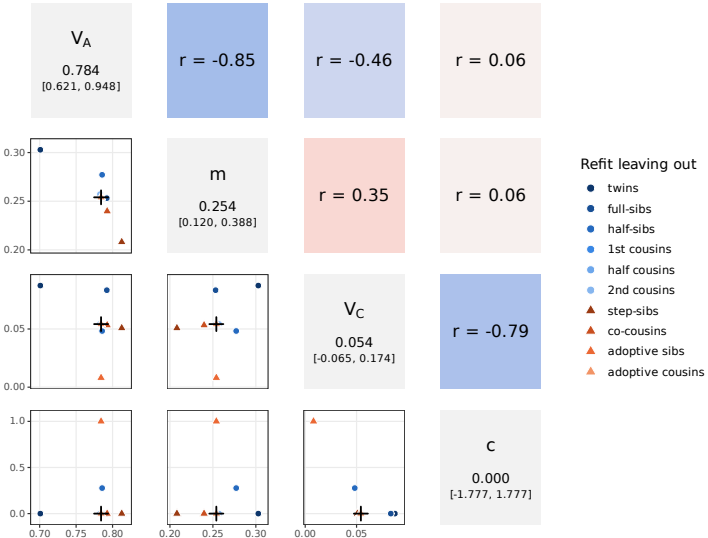
**Fit** (Marginal Type)

| | |
|---|---|
| $\chi^2/\mathrm{df}$ | 47.7 |
| CV RMSE | 0.071 |
| CV rank | 8 of 9 |

## Comments

- Since $c = 0$, the shared environment is shared by siblings only, with no shared environment beyond siblings.

Bars: predicted, by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line; arrow: truncated at 0 or 1).



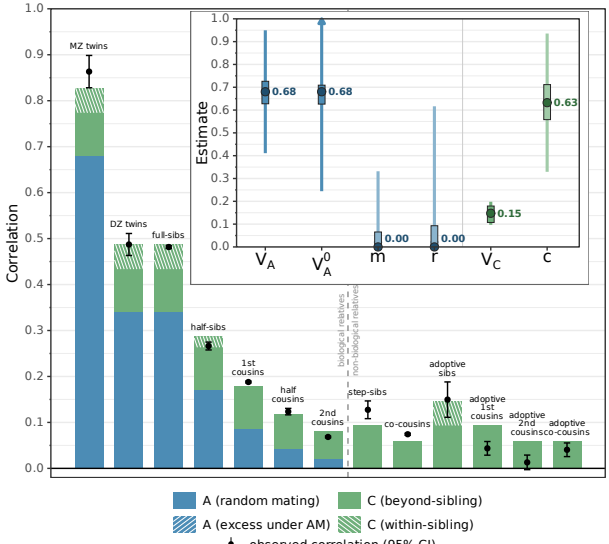

## Reading the plot

**Lower:** estimates from the 10 leave-one-type-out refits (colour: type left out); cross: full-data estimate.

**Upper:** $r$, the correlation of the refit estimates (jackknife sampling correlation).

**Diagonal:** estimate and Marginal Type CI.

With only 10 relative types, one or two influential types can set $r$: read it as the direction of the trade-off.




## Model

**A** and **C** fitted jointly.

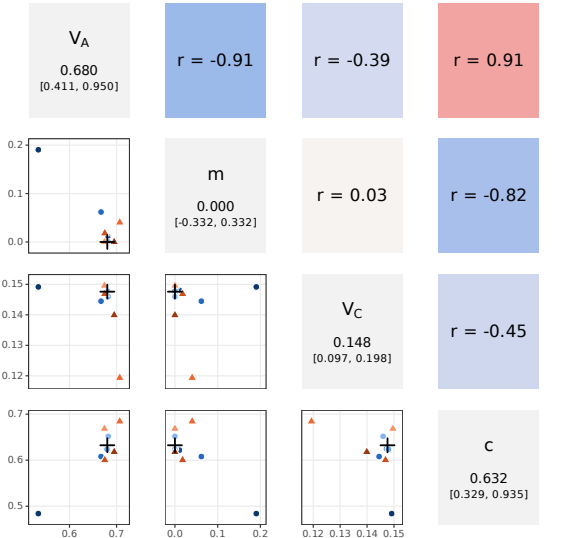
**Fit** (Marginal Type)

| | |
|---|---|
| $\chi^2/\mathrm{df}$ | 11.5 |
| CV RMSE | 0.050 |
| CV rank | 3 of 9 |

## Comments

- Finds no assortative mating ($m = r = 0$) on test performance, which is likely implausible.

Bars: predicted, by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line; arrow: truncated at 0 or 1).



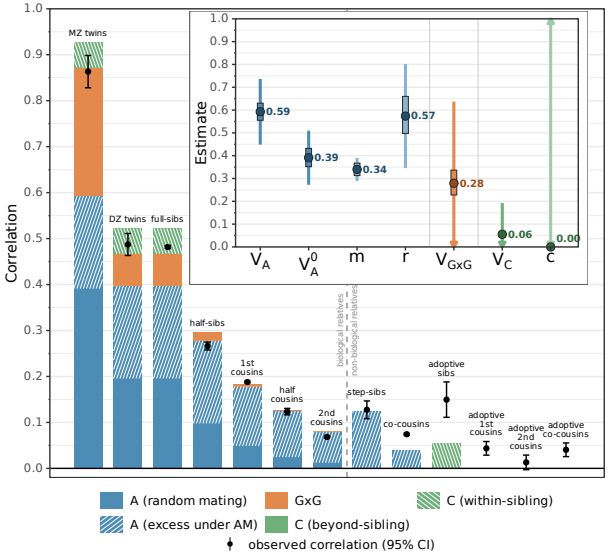

## Reading the plot

**Lower:** estimates from the 10 leave-one-type-out refits (colour: type left out); cross: full-data estimate.

**Upper:** $r$, the correlation of the refit estimates (jackknife sampling correlation).

**Diagonal:** estimate and Marginal Type CI.

With only 10 relative types, one or two influential types can set $r$: read it as the direction of the trade-off.




## Model

**A** and **GxG** fitted jointly first; **C** then fitted with their parameters held fixed.

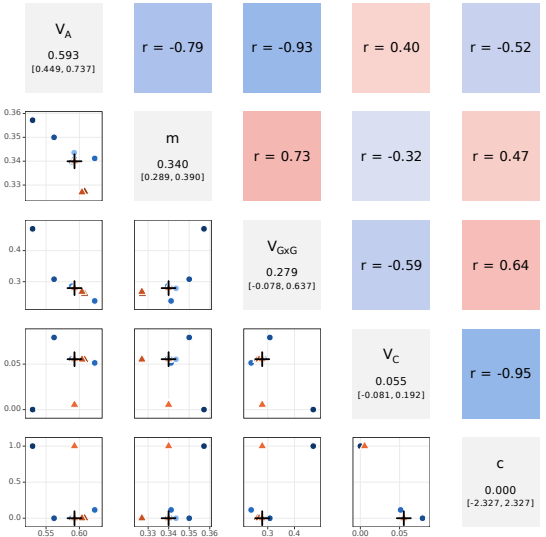
**Fit** (Marginal Type)

| | |
|---|---|
| $\chi^2/\mathrm{df}$ | 28.9 |
| CV RMSE | 0.065 |
| CV rank | 6 of 9 |

## Comments

- Like **A** → **C**, finds a shared environment between siblings only ($c = 0$).

Bars: predicted, by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line; arrow: truncated at 0 or 1).




Refit leaving out

- twins
- full-sibs
- half-sibs
- 1st cousins
- half cousins
- 2nd cousins
- step-sibs
- co-cousins
- adoptive sibs
- adoptive cousins

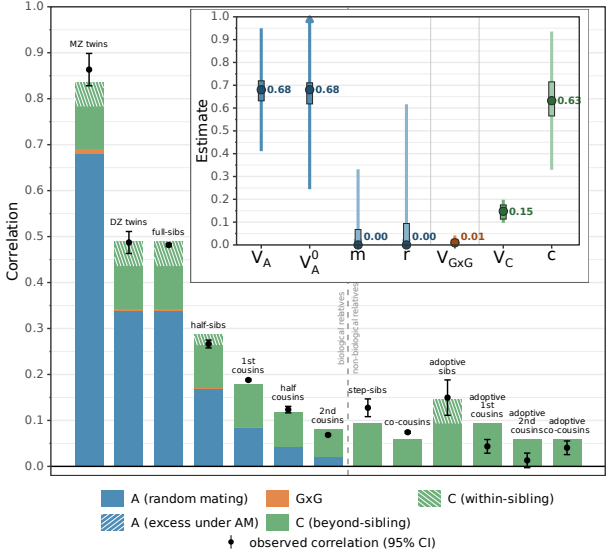
## Reading the plot

**Lower:** estimates from the 10 leave-one-type-out refits (colour: type left out); cross: full-data estimate.

**Upper:** $r$, the correlation of the refit estimates (jackknife sampling correlation).

**Diagonal:** estimate and Marginal Type CI.

With only 10 relative types, one or two influential types can set $r$: read it as the direction of the trade-off.




## Model

**A** and **C** fitted jointly first; **GxG** then fitted with their parameters held fixed.

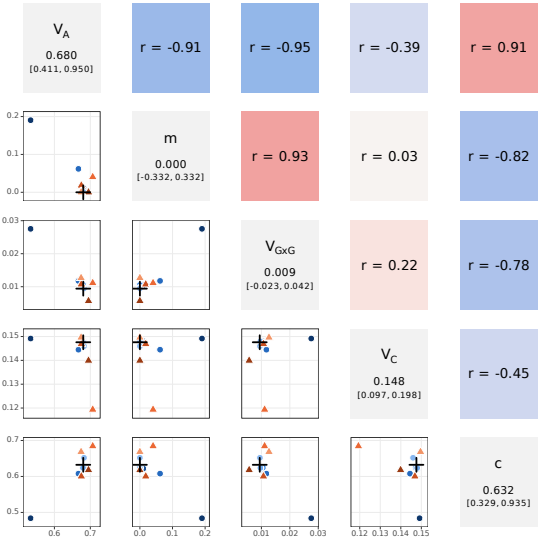
**Fit** (Marginal Type)

| | |
|---|---|
| $\chi^2/\mathrm{df}$ | 11.7 |
| CV RMSE | 0.046 |
| CV rank | 2 of 9 |

## Comments

- Like **A** + **C**, finds no assortative mating ($m = r = 0$).

Bars: predicted, by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line; arrow: truncated at 0 or 1).



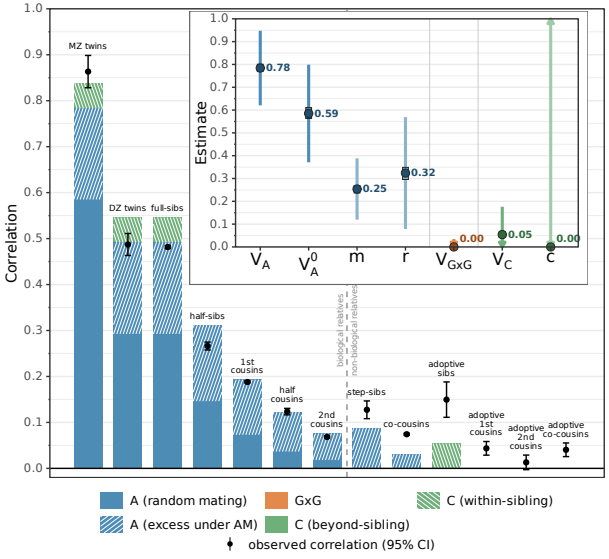

## Reading the plot

**Lower:** estimates from the 10 leave-one-type-out refits (colour: type left out); cross: full-data estimate.

**Upper:** $r$, the correlation of the refit estimates (jackknife sampling correlation).

**Diagonal:** estimate and Marginal Type CI.

With only 10 relative types, one or two influential types can set $r$: read it as the direction of the trade-off.




## Model

**A** fitted first; **GxG** and **C** then fitted jointly, with **A**'s parameters held fixed.

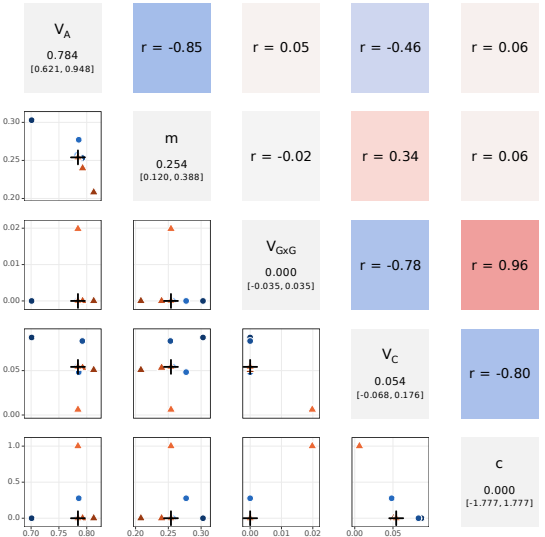
**Fit** (Marginal Type)

| | |
|---|---|
| $\chi^2/\mathrm{df}$ | 48.3 |
| CV RMSE | 0.072 |
| CV rank | 9 of 9 |

## Comments

- Like **A** → **C**, finds a shared environment between siblings only ($c = 0$).

Bars: predicted, by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line; arrow: truncated at 0 or 1).



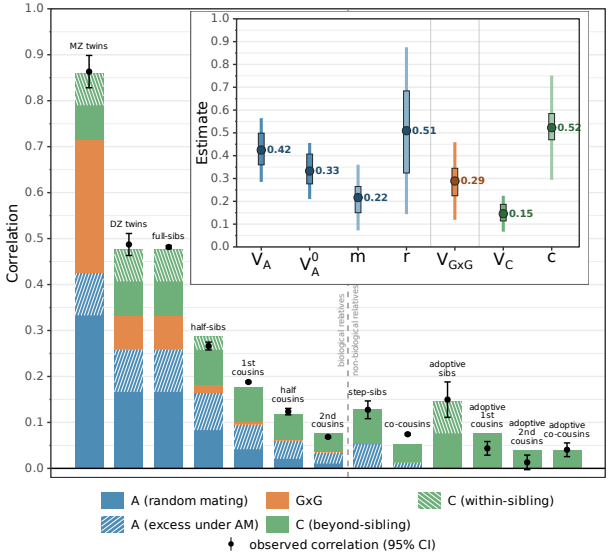

## Reading the plot

**Lower:** estimates from the 10 leave-one-type-out refits (colour: type left out); cross: full-data estimate.

**Upper:** $r$, the correlation of the refit estimates (jackknife sampling correlation).

**Diagonal:** estimate and Marginal Type CI.

With only 10 relative types, one or two influential types can set $r$: read it as the direction of the trade-off.



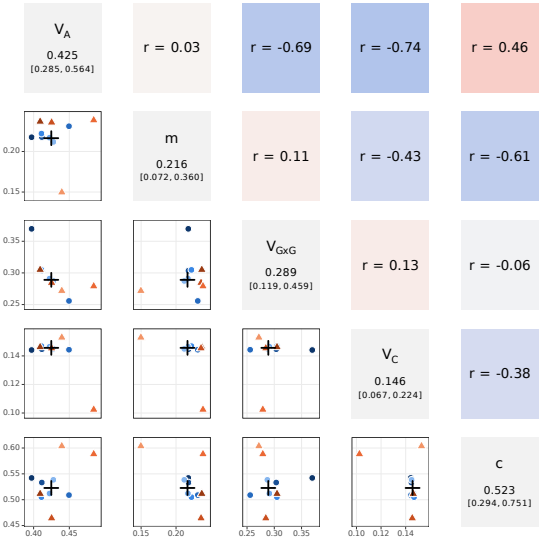

## Model

All three components fitted jointly. This is the example shown on the poster.

**Fit** (Marginal Type)

| | |
|---|---|
| $\chi^2/\mathrm{df}$ | 12.6 |
| CV RMSE | 0.035 |
| CV rank | 1 of 9 |

Bars: predicted, by component; black: observed (95% CI). Inset: estimate (circle), Marginal Pair CI (box), Marginal Type CI (line; arrow: truncated at 0 or 1).




Refit leaving out

- twins
- full-sibs
- half-sibs
- 1st cousins
- half cousins
- 2nd cousins
- step-sibs
- co-cousins
- adoptive sibs
- adoptive cousins

## Reading the plot

**Lower:** estimates from the 10 leave-one-type-out refits (colour: type left out); cross: full-data estimate.

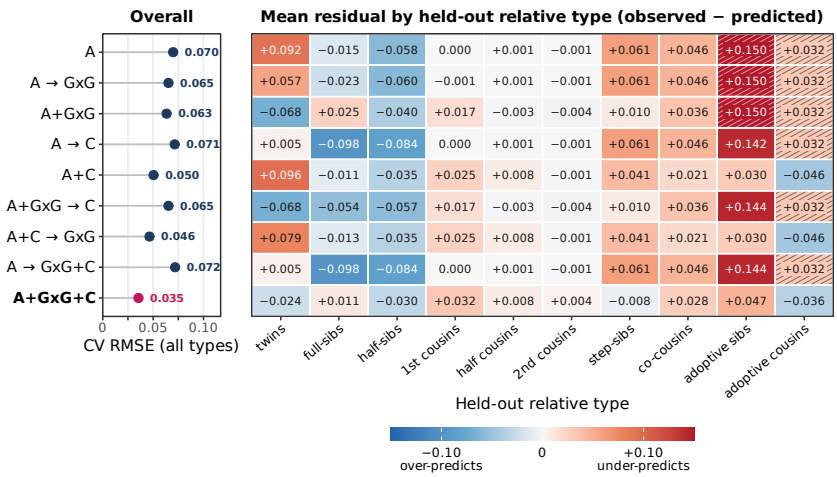
**Upper:** $r$, the correlation of the refit estimates (jackknife sampling correlation).

**Diagonal:** estimate and Marginal Type CI.

With only 10 relative types, one or two influential types can set $r$: read it as the direction of the trade-off.

# Leave-one-type-out cross-validation

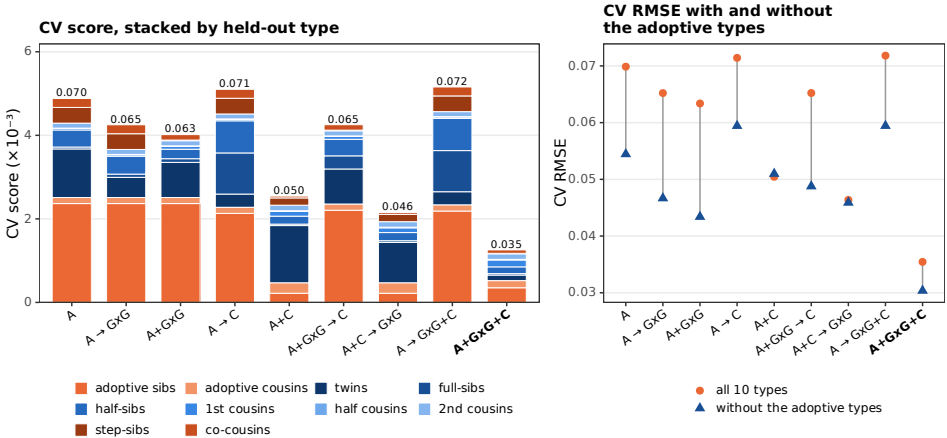
core models, Marginal Type fit




Each relative type is held out in turn and predicted from a fit to the other nine. Left: CV RMSE over all 10 types (precision-weighted; lower is better; best in pink). Right: each held-out type's mean residual, observed − predicted, precision-weighted as the CV score. Hatched: the model predicts 0 for the type, so its residual is the observed correlation. A residual near 0 can hide misses in both directions.

# Where the cross-validation error comes from

core models, Marginal Type fit




Left: each model's CV score (the precision-weighted squared error of the held-out correlations, averaged over the 10 relative types), split into each held-out type's share, with the adoptive types at the bottom. Label: CV RMSE. Right: CV RMSE over all 10 types and without the two adoptive types. Models without a jointly fitted **C** predict little or no similarity for adoptees.